Spectral indices and incidental amplitudes in pixel-based Bayesian component separation: tempered samplers, reference-frequency priors, and the polarized synchrotron index of the public WMAP and Planck maps

T. L. Svalheim

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1 Model, notation and the index targets in use

1.1 Data model

The data are maps d_b in bands $b = 1, \ldots, n_\mathrm{b}$, each a vector over elements p (pixels, and Stokes parameters in polarization), in thermodynamic units. The parametric model is

$$d_b = \sum_c m_{b,c}(\beta)\, a_c + n_b, \qquad n_b \sim \mathcal{N}(0, \mathsf{N}_b), \tag{1}$$

with one amplitude $a_c(p)$ per component c and element, a mixing coefficient $m_{b,c}$ that is the component spectrum averaged over the bandpass of band b, and spectral parameters β that are shared by many elements: one value for the sky, one per region, or a field at lower resolution than the amplitudes. Stacking bands and components, $d = \mathsf{M}(\beta)\, a + n$ with a of length k. We write

$$\mathsf{A} = \mathsf{M}^\mathsf{T}\mathsf{N}^{-1}\mathsf{M}, \qquad \hat{a} = \mathsf{A}^{-1}\mathsf{M}^\mathsf{T}\mathsf{N}^{-1}d, \qquad \mathsf{P} = \mathsf{N}^{-1} - \mathsf{N}^{-1}\mathsf{M}\mathsf{A}^{-1}\mathsf{M}^\mathsf{T}\mathsf{N}^{-1}, \tag{2}$$

all functions of β, so that

$$\chi^2(\beta, a) = \chi^2_\mathrm{min}(\beta) + (a - \hat{a})^\mathsf{T}\mathsf{A}\,(a - \hat{a}), \qquad \chi^2_\mathrm{min}(\beta) = d^\mathsf{T}\mathsf{P}d. \tag{3}$$

For the power-law component that carries the index, $m_b(\beta)$ is the band average of $(\nu/\nu_\mathrm{ref})^\beta$ in brightness temperature, converted to thermodynamic units, so that the amplitude is the brightness temperature at the reference frequency ν_ref. The reference frequency is a convention: it changes the meaning of a and not the model.

The amplitudes are what statisticians call incidental parameters (Neyman & Scott, 1948): their number grows with the data, and the parameters of interest, the indices, are few. Everything in this paper is a consequence of how the incidental parameters are removed.

1.2 Four ways to remove the amplitudes

Joint posterior with flat amplitude priors. With $p(a) \propto 1$ the joint posterior is $p(\beta)\exp[-\chi^2(\beta, a)/2]$ and integrating Eq. (3) over a gives the index marginal

$$\mathcal{L}_\mathrm{flat}(\beta) \propto \exp\!\left[-\tfrac{1}{2}\chi^2_\mathrm{min}(\beta)\right] \det \mathsf{A}(\beta)^{-1/2}. \tag{4}$$

A Gibbs sampler that draws $a \mid \beta$ with its fluctuation and then moves β on χ^2 at fixed a has exactly this marginal for β, whether or not it mixes well enough to show it (checked numerically and re-derived independently, E012). The determinant is the volume of amplitude space allowed at a given β. It is an implicit prior on the index and the subject of Sect. 3.

Profile likelihood. Maximizing over the amplitudes instead of integrating gives

$$\mathcal{L}_{\rm prof}(\beta) \propto \exp\left[-\tfrac{1}{2}\chi^2_{\rm min}(\beta)\right], \tag{5}$$

the spectral likelihood of Stompor et al. (2009) (their Eq. 9; their Eq. 10 is the marginal form). As a Bayesian target it is the marginal under the amplitude prior $p(a \mid \beta) \propto \det \mathsf{A}(\beta)^{1/2}$, a Jeffreys prior for the amplitude block conditional on the index. It is not the marginal under the joint Jeffreys prior of (a, β), which depends on a.

Gaussian amplitude prior. With $a \sim \mathcal{N}(\mu, \mathsf{S})$ the marginal keeps the form of Eq. (4) with $\mathsf{A} \to \mathsf{A} + \mathsf{S}^{-1}$ and the prior mean entering $\chi^2_{\rm min}$. We use two cases. A zero-mean prior with one free scale per index region, $a_{\rm s}(p) \sim \mathcal{N}(0, \tau_r^2)$ for p in region r, with τ_r maximized or marginalized under a prior flat in $\ln \tau_r$. And, for intensity, a prior whose mean and scale are tied to a template, with hyperparameters inferred. Under a change of reference frequency τ_r rescales by $(\nu'_{\rm ref}/\nu_{\rm ref})^{\beta_r}$, and both maximization and a log-flat prior are invariant under that rescaling, so the first case does not depend on the reference frequency.

Re-solving the amplitudes inside the index move. A fourth route is procedural: propose β', solve for new amplitudes at β', and accept on the χ^2 of the new pair. What this samples depends on how the new amplitudes are produced, and is the subject of Sect. 2.

1.3 Information and score

Let $\mathsf{M}_r = \partial \mathsf{M}/\partial \beta_r$ for index parameter r. For data that follow Eq. (1) with true amplitudes a, the profile score $s_r = -\frac{1}{2}\,\partial \chi^2_{\rm min}/\partial \beta_r$ has zero mean and covariance

$$\mathrm{Cov}(s_r, s_{r'}) = \mathsf{J}_{rr'} + \mathsf{T}_{rr'}, \qquad \mathsf{J}_{rr'} = (\mathsf{M}_r a)^{\mathsf{T}} \mathsf{P}\, (\mathsf{M}_{r'} a), \qquad \mathsf{T}_{rr'} = \mathrm{tr}\Big(\mathsf{A}^{-1} \mathsf{M}_r^{\mathsf{T}} \mathsf{P}\, \mathsf{M}_{r'}\Big), \tag{6}$$

while the expected curvature of $\frac{1}{2}\chi^2_{\rm min}$ is J alone (derivation in App. **??**). J is the Fisher information on the indices with the amplitudes profiled out. T does not depend on the signal: it is the part of the score variance that the noise in the fitted amplitudes contributes. For one power-law component, diagonal noise with weights $w_{pb} = 1/\sigma^2_{pb}$ and one region, these reduce to

$$\mathsf{J} = \sum_p u_p V_p, \qquad \mathsf{T} = \sum_p V_p, \qquad u_p = a_p^2 \sum_b w_{pb} m_b^2, \qquad V_p = \mathrm{Var}_{wm^2}(\ln \nu)_p, \tag{7}$$

with u_p the squared signal-to-noise ratio of amplitude p and V_p the variance of $\ln \nu$ over the bands under the weights $w_{pb} m_b^2$: the lever arm.

2 What the re-solve move converges to

2.1 The move

The move we analyse is natural, and a version of it was the spectral-index step of the per-pixel path in the development version of Commander4 that we started from (option `resolve_chisq` in our branch). Given the current (β, a):

1. propose β' from a symmetric random walk;
2. draw new amplitudes from their conditional at the proposed index, $a' \sim \mathcal{N}\big(\hat{a}(\beta'), \mathsf{A}(\beta')^{-1}\big)$;
3. accept (β', a') with probability $\min\{1, \exp[-(\chi^2(\beta', a') - \chi^2(\beta, a))/2]\}$.

The Gibbs sweep then continues with the amplitude group, which redraws $a \sim \mathcal{N}(\hat{a}(\beta), \mathsf{A}(\beta)^{-1})$, and possibly repeats the index move n times before that. The intent is to move the index and the amplitudes together and so avoid the slow mixing of the conditional index step. Write $X = (a - \hat{a})^{\mathsf{T}} \mathsf{A} (a - \hat{a})$, so that $\chi^2 = \chi^2_{\rm min}(\beta) + X$ by Eq. (3). For an amplitude draw from the conditional, $X \sim \chi^2_k$ whatever β is.

2.2 Alone, the move is exact for the profile posterior

Because $\mathrm{E}[\exp(-X/2)] = 2^{-k/2}$ does not depend on β, the quantity $\exp[-\chi^2(\beta, a)/2]$ at a fresh conditional draw is an unbiased estimator of $\mathcal{L}_{\mathrm{prof}}(\beta)$ up to a constant. Steps 1 to 3, with the accepted amplitudes kept as part of the state, are therefore a pseudo-marginal Metropolis-Hastings kernel (Beaumont, 2003; Andrieu & Roberts, 2009). Its extended target is

$$\pi(\beta, a) \propto p(\beta)\, \exp\!\big[-\tfrac{1}{2}\chi^2_{\min}(\beta)\big]\, \det \mathsf{A}(\beta)^{1/2} \exp\!\Big[-(a-\hat{a})^{\mathsf{T}} \mathsf{A}\, (a-\hat{a})\Big], \tag{8}$$

whose index marginal is the profile posterior and whose amplitude conditional is $\mathcal{N}(\hat{a}, (2\mathsf{A})^{-1})$: the amplitudes carry half their posterior covariance. So the move does have a target, the target is the profile posterior, and the determinant never enters. This is why chains run with this move have means close to the profile posterior. Under Eq. (8), $X \sim \mathrm{Gamma}(k/2, 1)$ with mean $k/2$.

2.3 The amplitude step of the Gibbs sweep breaks it

The amplitude group redraws a with the full conditional covariance A^{-1}, which resets X to a χ^2_k variate with mean k. That is the conditional of the flat-prior joint posterior and not of Eq. (8). The two kernels share no invariant distribution. With one index move per amplitude draw, both the current and the proposed estimate are fresh at every index move, and the composite is Monte Carlo within Metropolis in the terminology of Andrieu & Roberts (2009), following O'Neill et al. (2000) and Beaumont (2003). Andrieu & Roberts (2009) note that the target is typically not the invariant density of such a chain and that the existence of an invariant distribution is not obvious in general. Here it can be written down.

2.4 The stationary law: a tempered profile posterior

With one move per amplitude draw the index chain is Markov on its own, with acceptance probability

$$\alpha(D) = \mathrm{E}\big[\min\{1, e^{-(D+Z)/2}\}\big], \qquad D = \chi^2_{\min}(\beta') - \chi^2_{\min}(\beta) - 2\ln\frac{p(\beta')}{p(\beta)}, \qquad Z = X' - X, \tag{9}$$

where X and X' are independent χ^2_k variates, so Z is symmetric with variance $4k$. In the limit of small proposal steps the chain is a diffusion with drift proportional to $\alpha'(0)$ and diffusion coefficient proportional to $\alpha(0)$, and its stationary density is

$$\pi(\beta) \propto \big\{p(\beta)\exp\!\big[-\tfrac{1}{2}\chi^2_{\min}(\beta)\big]\big\}^{1/T}, \qquad T = \frac{1}{2} + \frac{1}{4c_k}, \qquad c_k = \mathrm{E}\big[e^{-Z/2};\, Z>0\big] = 2^{-(k+1)}\binom{k}{k/2}. \tag{10}$$

The formula is exact for every k and every shape of $\chi^2_{\min}$ in the small-step limit; $T(0) = 1$ recovers Metropolis. For large k,

$$T = \sqrt{\frac{\pi k}{8}} + \frac{1}{2} + O(k^{-1/2}), \tag{11}$$

so the index posterior is the profile posterior with its width multiplied by $\sqrt{T} \simeq 0.79\, k^{1/4}$. The stationary acceptance rate is $\frac{1}{2} + c_k$, close to one half and independent of the step size, which is why such a chain looks healthy.

Equation (10) does not need χ^2_k noise. For any symmetric acceptance noise Z with density f_Z, $T \simeq \frac{1}{2} + 1/(8 f_Z(0))$, and for Gaussian noise

$$T \simeq \frac{1}{2} + \sqrt{\frac{\pi}{8}}\, \sigma_E, \qquad \sigma_E = \tfrac{1}{2}\,\mathrm{sd}(Z), \tag{12}$$

with σ_E the standard deviation of the noise in the estimated energy difference, equal to $1/\sqrt{2}$ times the standard deviation of the χ^2 at a fresh amplitude draw. The leading term is the

Table 1: Variance of the stationary index distribution of the composite chain in units of the profile posterior variance, one index move per amplitude draw, for $\chi^2_{\min} = \beta^2$ and exact χ^2_k refreshes. Columns: Eq. (10); its leading term; the stationary vector of the exact kernel on a grid of 1201 points, with the acceptance function computed by convolution of the χ^2_k density; and a simulation of 10^4 chains with a proposal step of $0.1\sqrt{T}$ (E035, independent review). The grid and the simulation sit 0.1 to 0.2 percent below the law because of the finite step. Controls: with no refresh, or with a refresh from Gamma$(k/2, 1)$, the variance is 1.000 ± 0.001 at $k = 4$, 8 and 16.

k	T, Eq. (10)	$\sqrt{\pi k/8}$	exact kernel	simulation	acceptance
10	2.532	1.982	2.524	2.525 ± 0.008	0.621
100	6.782	6.267	6.769	6.751 ± 0.024	0.538
1 000	20.32	19.82	20.29	20.26 ± 0.07	0.511
10 000	63.17	62.67	63.07	63.30 ± 0.23	0.503
100 000	198.67	198.17	198.35	199.39 ± 0.72	0.500
200 000	280.75	280.25	280.31	280.22 ± 1.01	0.500

effective temperature that Ball et al. (2003) derived for the sign rule (accept if and only if the noisy energy difference is negative; their Eq. 7), and that Bardenet et al. (2017) recover without assuming Gaussian noise (their Sect. 6.3). The Metropolis rule on the same noisy difference adds the $\frac{1}{2}$. Equation (12) is the form to use when the stored amplitudes are not the drawn ones, as below.

Several index moves per amplitude draw. After the first accepted move the retained X is no longer fresh: it is biased low, and the next proposal has to beat it. In the large-k limit the accepted moves are the lower records of the sequence of fresh X values, the j-th proposal after a redraw is accepted with probability $1/(j+1)$, and the stationary law remains a power of the profile posterior with temperature $T_n = R_n T_1$,

$$R_n = \frac{H_{n+1} - 1}{\sqrt{\pi}\, N_\varphi(n)}, \qquad N_\varphi(n) = \sum_{j=1}^{n} j \int \varphi(t)^2 \, [1 - \Phi(t)]^{j-1} \, \mathrm{d}t, \tag{13}$$

where H_n is the harmonic number, φ and Φ are the standard normal density and distribution function, and $N_\varphi(n)$ is the sum over proposals of the expected density at the running minimum. $R_2 = 5/6$ exactly, and numerically $R_n = 0.833$, 0.647, 0.543, 0.467 and 0.357 for $n = 2$, 5, 10, 20 and 100: more index moves narrow the law only as $(\ln n)^{-1/4}$ in width, and the $k^{1/4}$ scaling is untouched.

The repair that is exact and useless. Redrawing the amplitudes with half their covariance, $a \sim \mathcal{N}(\hat{a}, (2\mathsf{A})^{-1})$, is the conditional of Eq. (8) and makes the composite exact for the profile posterior. Its acceptance rate at fixed index is $2I_{1/3}(k/2, k/2)$, with I the regularized incomplete beta function, which decays as $e^{-0.059k}$: 6×10^{-4} at $k = 100$. The useful repairs evaluate Eq. (5) or Eq. (4) directly in the acceptance, which costs one extra log-determinant per proposal at most (Sect. **??**).

2.5 Tests

Table 1 and Fig. 1 compare Eq. (10) with the exact kernel and with simulated chains. The law was derived twice, the second time by an independent reviewer who was given the algorithm and not our derivation. The two derivations agree, the reviewer's exact-kernel calculation reproduces every row of our own toy within 1.4 standard errors, and the controls return the profile posterior exactly, so the excess is caused by the full-covariance redraw and by nothing else. For proposal steps that are not small the kernel is no longer reversible with respect to a power of the profile

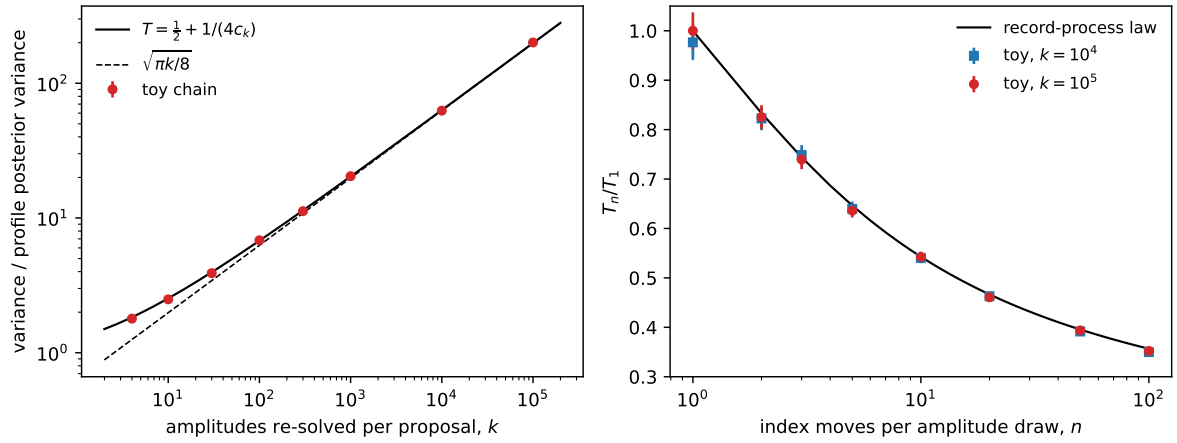


Figure 1: The stationary law of the re-solve move inside a Gibbs sweep. Left: temperature of the tempered profile posterior against the number of amplitudes re-solved per proposal, for one index move per amplitude draw: Eq. (10) (line), its leading term (dashed) and the toy chain (points). Right: the ratio T_n/T_1 for n index moves per amplitude draw: the record-process law of Eq. (13) (line) and the toy at $k = 10^4$ and 10^5 (points). [figure from `paperA1_fig_temper.py`]

posterior, but the variance stays within 8 percent of T at any step size (0.918 T for steps of four stationary widths or more).

The same toy reproduces, within 5 percent, four earlier measurements of the width of chains run with this move on simulated maps, with k from 450 to 3072 and one or four moves per amplitude draw (measured 3.9, 5.7, 3.2 and 4.4 profile widths; toy 4.0 to 4.1, 5.7, 3.0 and 4.3; E002, E005).

Real maps. On nside-64 intensity maps (17 bands from Haslam to Planck 857 GHz, 38 165 unmasked pixels, five components, so $k = 190\,825$; 20 index moves per amplitude draw) two chains gave widths of 17.0 and 16.4 profile widths for the synchrotron index and 16.4 and 16.6 for the dust index (E014). The law with exact stored draws predicts 13.1 to 13.6 at these settings. The difference has a specific cause. This code path stores the per-pixel amplitudes as spherical-harmonic coefficients to $\ell_{\max} = 3N_{\rm side} - 1$, and the χ^2 in the acceptance is evaluated on the round-tripped maps. The round trip is not exact on the HEALPix grid, so the stored amplitudes differ from the drawn ones. The log of the run records the log acceptance ratio of every proposal, and at the first move of each iteration both amplitude sets are fresh: the scatter of those 3200 values is 682 ± 9 where exact draws give $\sqrt{k} = 437$, a factor 2.44 ± 0.06 in variance. With that one measured number in Eq. (12) the toy gives widths of 15.9 and 16.6 and an acceptance rate of 0.075, against the measured 0.074 and 0.077. The storage residual also has a gradient in the dust index, which predicts a shift of the chain mean of +3.1 profile widths where $+3.0 \pm 0.5$ was measured. A run of the same configuration with the amplitudes stored as maps is predicted to give 13.1 and 13.6 widths and no shift; [that run is not yet done].

2.6 What this means for existing chains

A chain run with this move reports index uncertainties that are too wide by $\sqrt{T}$, a factor of order ten at map-level k, with a mean close to the profile posterior. It does not sample the flat-prior joint posterior that the rest of the Gibbs sweep assumes. The statement applies to the move as defined in steps 1 to 3 with exact conditional draws. Samplers that propose the index and then solve for the conditional mean, or that evaluate a marginal likelihood in the acceptance, are different algorithms. Commander3's index samplers are of the latter kinds (we read the code and did not run it).

3 The flat amplitude prior, written in terms of the reference frequency

That the flat-prior marginal of Eq. (4) pulls spectral indices is known. Eriksen et al. (2008) found a global offset biased by more than 4σ under uniform priors and removed it with a Jeffreys prior on the index. Dunkley et al. (2009), fitting one polarized synchrotron index per $N_{\rm side} = 2$ pixel to the five-year WMAP maps with uniform priors on the CMB and synchrotron amplitudes defined at 23 GHz, found index posteriors centred on -2.6 in simulations, noted that the effect grows with the number of pixels that share an index and that a Jeffreys prior on the index does not remove it, and divided their prior by a factor estimated from a noise-only simulation. Alonso et al. (2017) cancel the determinant with a "volume prior", which makes their target the profile likelihood, and show biases of more than 3σ without it. What has not been written down, as far as we can find, is how large the pull is in closed form and what sets its sign. Both follow from one identity.

3.1 An exact identity

Change the reference frequency of the power-law component from $\nu_{\rm ref}$ to $\nu'_{\rm ref}$. The mixing column of every amplitude in index region r is multiplied by $(\nu_{\rm ref}/\nu'_{\rm ref})^{\beta_r}$, so $\mathsf{M}' = \mathsf{M}\,\mathsf{D}$ with D diagonal, $\chi^2_{\rm min}$ is unchanged, and $\det \mathsf{A}' = \det \mathsf{A} \prod_r (\nu_{\rm ref}/\nu'_{\rm ref})^{2 n_r \beta_r}$, with n_r the number of power-law amplitudes in region r. Therefore

$$\ln \mathcal{L}_{\rm flat}(\beta; \nu'_{\rm ref}) = \ln \mathcal{L}_{\rm flat}(\beta; \nu_{\rm ref}) + \sum_r n_r\, \beta_r \ln \frac{\nu'_{\rm ref}}{\nu_{\rm ref}}, \tag{14}$$

for any number of components, any bandpasses and any noise covariance. Stating a flat amplitude prior at a different frequency is the same as multiplying the index prior by an exponential, $e^{\lambda\beta}$ with $\lambda = n \ln(\nu'_{\rm ref}/\nu_{\rm ref})$: 34 e-folds per unit of index for 128 amplitudes moved from 23 to 30 GHz. The profile likelihood and the zero-mean prior with a free scale do not change.

3.2 The shift and its pivot

To first order the maximum of $\mathcal{L}_{\rm flat}$ sits at the profile maximum plus $\delta\beta = \mathsf{J}^{-1} g$, with $g_r = -\frac{1}{2}\,\partial \ln\det \mathsf{A}/\partial\beta_r$ the gradient of the volume term. For one power-law component with delta bands and diagonal noise, $-\frac{1}{2}\ln A_p = \beta \ln \nu_{\rm ref} - \frac{1}{2}\ln \sum_b w_{pb}\, \nu_b^{2\beta}$, so

$$g = \sum_p \ln \frac{\nu_{\rm ref}}{\bar\nu_p}, \qquad \ln \bar\nu_p = \frac{\sum_b w_{pb}\, \nu_b^{2\beta} \ln \nu_b}{\sum_b w_{pb}\, \nu_b^{2\beta}}, \tag{15}$$

where $\bar\nu_p$ is the frequency at which amplitude p is best measured: the geometric mean of the band frequencies under the weights (inverse noise variance times squared spectrum) that also define the lever arm V_p in Eq. (7). When the amplitudes of a region share a noise pattern, $\bar\nu_p = \nu_{\rm piv}$ and, with $\sigma_\beta^2 = 1/\mathsf{J}$ the profile variance of the region index,

$$\boxed{\delta\beta = \ln\!\left(\frac{\nu_{\rm ref}}{\nu_{\rm piv}}\right) n\, \sigma_\beta^2 = \ln\!\left(\frac{\nu_{\rm ref}}{\nu_{\rm piv}}\right) \sigma_1^2} \tag{16}$$

where $\sigma_1^2 = n\sigma_\beta^2$ is the index variance that a single amplitude carries. Three things follow.

The shift does not average down. Adding amplitudes at fixed depth leaves σ_1 unchanged, so $\delta\beta$ stays the same while the posterior width falls as $n^{-1/2}$. In units of the posterior width the pull grows as $\sqrt{n}$: this is the observation of Dunkley et al. (2009) that the effect increases with the number of pixels per index, and it is the Neyman-Scott structure (Neyman & Scott, 1948). More data of the same kind make it worse.

Table 2: The flat-prior marginal against Eq. (16) on simulations at the white-noise depth of the WMAP and LFI polarization maps (E036). One index per $N_{\rm side} = 2$ pixel (128 Q/U amplitudes at $N_{\rm side} = 16$), exact one-dimensional posteriors on a grid, 100 noise realizations; truth: the Cosmoglobe DR1 polarized synchrotron map smoothed to 5° with $\beta = -3.1$. Regions are grouped by their profile index error σ_β. "Shift" is the mean of posterior mean minus truth. The predicted value is Eq. (16) with the class median of σ_β (in brackets) and the pivot of the band set; it is first order in the shift and uses one number for a class, so it degrades where the shift exceeds the posterior width.

model	σ_β class (median)	reference	shift	predicted	shift / posteri
synchrotron only,	below 0.03 (median 0.019)	23 GHz	−0.005	−0.006	−
bands K, 30, Ka, Q, 44		30 GHz	+0.007	+0.006	+
($\nu_{\rm piv} = 26.4$ GHz)		60 GHz	+0.037	+0.038	
	0.03 to 0.1 (0.059)	23 GHz	−0.061	−0.062	−
		30 GHz	+0.062	+0.057	+
		60 GHz	+0.341	+0.37	
	0.1 to 0.25 (0.161)	23 GHz	−0.40	−0.46	
		30 GHz	+0.34	+0.42	
free CMB and dust, flat priors,	0.03 to 0.1	23 GHz	+0.006		
adding V, 70, 353 GHz		30 GHz	+0.167		
	0.1 to 0.25	23 GHz	+0.058		
		30 GHz	+0.859		
the same with a CMB prior	0.03 to 0.1	23 GHz	−0.060		
of 0.2 μK per element		30 GHz	+0.063		

The sign is set by the convention. The index is pulled steep if the reference lies below the pivot and flat if above. A flat prior is neutral only when stated at the pivot.

The pivot depends on the other components and on their priors. A component with a free amplitude and a spectrum that does not depend on the index (the CMB, dust with fixed parameters) contributes nothing to g directly, but projecting it out replaces the weights w_{pb} by the projected weight matrix $\tilde{\mathsf{W}}_p = \mathsf{W}_p - \mathsf{W}_p\mathsf{F}(\mathsf{F}^\mathsf{T}\mathsf{W}_p\mathsf{F} + \mathsf{S}^{-1})^{-1}\mathsf{F}^\mathsf{T}\mathsf{W}_p$, with F the spectra of the other components and S their prior covariance. The pivot becomes a generalized mean under $\tilde{\mathsf{W}}_p$, which has negative off-diagonal elements and can lie outside the range of the bands. With a free CMB amplitude under a flat prior the index information sits in the lowest band and the pivot moves to or below it. With a CMB prior the size of the signal it moves back.

For several regions, several components and dense noise, g and J are matrices and Eq. (14) gives the exact difference between two references, $\delta\beta(\nu'_{\rm ref}) - \delta\beta(\nu_{\rm ref}) \simeq \mathsf{J}^{-1}\,[\,n_r \ln(\nu'_{\rm ref}/\nu_{\rm ref})\,]$, a coupled form that we test on real data in Sect. **??**.

3.3 Sizes at the depth of the public polarization data

Table 2 gives the sizes on a realistic polarized sky at the white-noise depth of the WMAP and LFI maps. For the synchrotron-only model the pivot of the five lowest bands is 26.4 GHz, and references of 23 and 30 GHz, the two in use, sit on opposite sides of it: the same data give an index steeper than the truth by 0.06 in one convention and flatter by 0.06 in the other, one posterior standard deviation each way, for regions measured to 0.06. For regions measured to 0.14 the pull is 0.4 and 0.3. The difference between the two references follows Eq. (14): $\ln(30/23) \times 128 \times 0.059^2 = 0.118$ against a measured 0.123. Doubling and quadrupling the noise leaves the shift at a given σ_β unchanged, as Eq. (16) requires (−0.063 and −0.073 at 23 GHz for the class 0.03 to 0.1; E036). The profile likelihood on the same simulations is unbiased to 0.002 in the first class and 0.02 in the second (Sect. **??**).

With free CMB and dust amplitudes under flat priors the pivot moves to 22 to 23 GHz. A

flat prior stated at 23 GHz is then nearly neutral and one stated at 30 GHz pulls the index flat by 0.17 to 0.86. With a Gaussian CMB prior of 0.2 μK per element, the size of the CMB polarization signal at this resolution, the pivot and the shifts return to the synchrotron-only values, and so does the index error, which the flat CMB prior had inflated by a factor 1.63. The prior on the CMB amplitude matters more for the synchrotron index than the prior on the synchrotron amplitude does, as long as the latter is not flat.

An existing real-data comparison. An earlier experiment of ours (E022) evaluated both the flat-prior marginal and the profile likelihood on real WMAP Ka to V, LFI and 353 GHz polarization maps in the regions of Svalheim et al. (2023), at a 30 GHz reference, for two noise models. Equation (16) says the difference of the two estimates divided by $n\sigma_\beta^2$ is $\ln(30/\nu_{\rm piv})$ whatever the noise model. In the Spur region the two noise models give shifts of +0.271 and +0.172 for $n\sigma_\beta^2 = 1.913$ and 1.216: a pivot of 26.04 GHz in both cases, to 0.2 percent. With a CMB prior of 0.19 μK the implied pivot is 29.5 to 30.2 GHz. So for that band set, with the CMB constrained, a 30 GHz reference happens to sit at the pivot and a flat synchrotron prior is neutral, which may be why the index posteriors of Svalheim et al. (2023) show no such pull. With K band added the pivot moves to about 26 GHz and a 30 GHz reference is no longer neutral. Sect. **??** tests the identity directly on the WMAP maps with their dense covariances. [section to be written] [section to be written] [section to be written] [section to be written] [section to be written]
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References

Alonso, D., Dunkley, J., Naess, S., & Thorne, B. 2017, Phys. Rev. D, 95, 043504 (arXiv:1608.00551)

Andrieu, C., & Roberts, G. O. 2009, Ann. Statist., 37, 697 (arXiv:0903.5480)

Ball, R. C., Fink, T. M. A., & Bowler, N. E. 2003, Phys. Rev. Lett., 91, 030201 (arXiv:cond-mat/0301179)

Bardenet, R., Doucet, A., & Holmes, C. 2017, J. Mach. Learn. Res., 18, 1 (arXiv:1505.02827)

Beaumont, M. A. 2003, Genetics, 164, 1139

Bennett, C. L., Larson, D., Weiland, J. L., et al. 2013, ApJS, 208, 20

Dunkley, J., Spergel, D. N., Komatsu, E., et al. 2009, ApJ, 701, 1804 (arXiv:0811.4280)

Eriksen, H. K., Jewell, J. B., Dickinson, C., et al. 2008, ApJ, 676, 10 (arXiv:0709.1058)

Fuskeland, U., Wehus, I. K., Eriksen, H. K., & Næss, S. K. 2014, ApJ, 790, 104 (arXiv:1404.5323)

Huber, P. J. 1967, in Proc. Fifth Berkeley Symp. Math. Statist. Probab., Vol. 1, 221

Neyman, J., & Scott, E. L. 1948, Econometrica, 16, 1

O'Neill, P. D., Balding, D. J., Becker, N. G., Eerola, M., & Mollison, D. 2000, Appl. Statist., 49, 517

Planck Collaboration Int. LVII. 2020, A&A, 643, A42

Planck Collaboration XI. 2020, A&A, 641, A11

Stompor, R., Leach, S., Stivoli, F., & Baccigalupi, C. 2009, MNRAS, 392, 216 (arXiv:0804.2645)

Svalheim, T. L., Andersen, K. J., Aurlien, R., et al. 2023, A&A, 675, A14 (arXiv:2011.08503)

Watts, D. J., Basyrov, A., Eskilt, J. R., et al. 2023, A&A, 679, A143 (arXiv:2303.08095)

Weiland, J. L., Addison, G. E., Bennett, C. L., Halpern, M., & Hinshaw, G. 2022, ApJ, 936, 24 (arXiv:2203.11445)

White, H. 1982, Econometrica, 50, 1